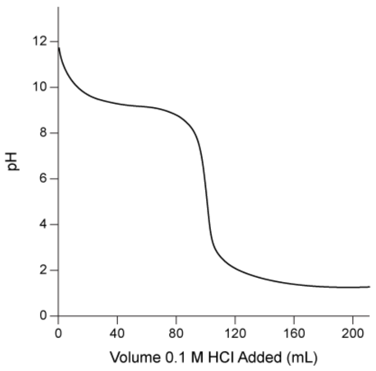
Consider the titration curve for aqueous ammonia shown below.

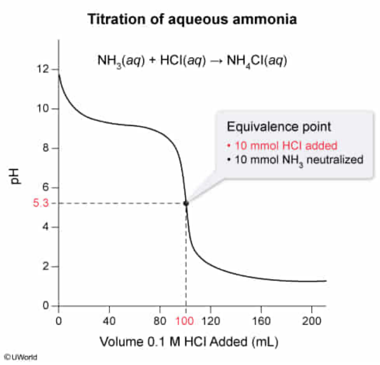


The quantity of hydrochloric acid required to reach the equivalence point in the titration is:

- A. 9 mmol. [8%]
- B. 10 mmol. [85%]
- C. 12 mmol. [4%]
- D. 16 mmol. [1%]

Correct
85% answered correctly

Explanation:



In an acid-base titration, a measured amount of an acid (or a base) solution (titrant) with a known concentration is slowly added to another solution containing a base (or an acid) of an unknown concentration to be determined. The acid (or base) being titrated is fully neutralized at the **equivalence point**, where the number of equivalents of acid exactly equals the number of equivalents of base.

The pH changes rapidly when passing through the equivalence point. On a titration curve, this is seen as a nearly vertical segment, and the equivalence point is located at the middle of this **vertical segment**. The quantity (moles) of acid or base delivered at this point is calculated by **multiplying** the **volume of titrant** added by the titrant **concentration**.

In this titration, 100 mL of 0.1 M HCl were required to reach the equivalence point. Accordingly, the number of millimoles of HCl required to reach this point is

$$100 \qquad \text{mL} \qquad \times$$

Because each mole of ammonia gives one equivalent of base and each mole of HCl gives one equivalent of acid, the neutralization reaction in this titration proceeds with a 1:1 molar ratio. Therefore, 10 mmol of HCl were required to neutralize 10 mmol of NH_3 present in the unknown ammonia solution.

(Choices A, C, and D) Quantities of 9 mmol, 12 mmol, and 16 mmol result from adding 90 mL, 120 mL, and 160 mL of 0.1 M HCl, respectively, to the ammonia solution. These volumes of titrant are less than or greater than the volume required to reach the equivalence point.

Educational objective:
The equivalence point in an acid-base titration occurs when the number of equivalents of acid equals the number of equivalents of base; it is seen as the midpoint of a nearly vertical segment of the titration curve. The volume of titrant at the equivalence point multiplied by its concentration gives the moles of titrant species added.

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